Principle of Thermodynamics

Bohaz

15th August 2026

Contents

1 The first Law **2**

2 The Second Law **8**

3 Free Energy **13**

1 The first Law

In the previous series, we introduced the basic principles of quantum mechanics—the theory that describes atoms and molecules. Thermodynamics, by contrast, describes matter on a macroscopic scale. Remarkably, it was developed before quantum mechanics, at a time when even the existence of atoms was still debated. Yet, guided by the principles of thermodynamics, engineers were already able to design increasingly efficient steam engines.

To study a physical system, we first need a way to describe its state. In quantum mechanics, the state is represented by a state vector, $|\psi\rangle$. In thermodynamics, an equilibrium state is characterized by macroscopic variables related through an equation of state (EOS). Fortunately, we are already familiar with one important example: the ideal gas law,

$$pV = nRT,$$

where p is the pressure, V is the volume, n is the amount of gas in moles, T is the absolute temperature in kelvin, and

$$R = 8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$$

is the ideal gas constant.

If the amount of gas is fixed, only two of the three variables p, V, and T can be chosen independently. For example, once V and T are specified, the pressure is determined by the ideal gas law. Thus, the equilibrium state of this simple system can be specified by just two variables, such as (V, T).

This description is remarkably compact. It is far simpler than specifying the infinitely many amplitudes that may be required to describe a quantum state, or tracking the positions and momenta of every atom in the gas.

We now introduce two central concepts: the 'system' and the 'surroundings'. The system is the part of the universe we choose to study, while everything else is the surroundings. The surface separating them is called the "boundary".

Consider a solution in a test tube. An open test tube is an "open system", able to exchange both matter and energy with its surroundings. Once sealed, it becomes a "closed system": it can still exchange energy, but not matter. Because the test tube is rigid, its volume is approximately constant, so pressure–volume work is negligible. To allow such work, we instead need a movable boundary, such as a piston. A sealed and perfectly insulated system is "isolated": it exchanges neither matter nor energy with its surroundings.

Energy crosses the boundary as either "heat", q, or "work", w. Heat is energy transfer down temperature gradient, while work is force time distance.

We take energy transferred into the system as positive. Thus, $q > 0$ when the system absorbs heat, and $w > 0$ when work is done on it.

Heat and work are not energy stored in the system; they are two ways of transferring energy across its boundary.

What we have described is the 'first law of thermodynamics':

$$\Delta U = q + w. \tag{1.1}$$

It states that the internal energy U of a system can change through either heat or work. Once energy enters the system, it does not retain a label indicating how it arrived—just

as a purchase may be made with cash or a credit card, but the money has the same value either way.

Notice that we write ΔU, but not Δq or Δw. Internal energy is stored in the system, so it makes sense to speak of a change in U. Heat and work, however, are not stored: they describe energy being transferred across the system boundary. Therefore, we write q and w, rather than "changes" in heat or work. It follows directly, that for an isolated system, there is no heat transfer to, or work done on the system, so energy of an isolated system is conserved.

We now consider some important properties of the internal energy U. First, U is a **state function**: its value depends only on the state of the system. For a system with a fixed amount of matter, we may write, for example,

$$U \equiv U(V, T). \tag{1.2}$$

Each state (V, T) can be represented as a point in a two-dimensional **state space**, and U assigns a value to every point. The change in internal energy between an initial state (V_1, T_1) and a final state (V_2, T_2) is

$$\Delta U = \int_{V_1,T_1}^{V_2,T_2} \mathrm{d}U = U(V_2, T_2) - U(V_1, T_1). \tag{1.3}$$

Because U is a state function, ΔU depends only on the initial and final states, not on the path taken between them. Importantly, the change in a state function is **path independent**. For example, we may first heat a gas and then compress it, or first compress it and then heat it. Provided that the initial and final states are the same, the change in internal energy is also the same. Thus, the integral of $\mathrm{d}U$ is independent of the path between the two states.

It follows immediately that

$$\oint \mathrm{d}U = 0, \tag{1.4}$$

where the circle on the integral sign indicates integration around a closed path. After returning to the initial state, the internal energy must return to its initial value.

The quantity $\mathrm{d}U$ is therefore called an **exact differential**. Since $U = U(V, T)$, it can be written as

$$\mathrm{d}U = \left(\frac{\partial U}{\partial V}\right)_T \mathrm{d}V + \left(\frac{\partial U}{\partial T}\right)_V \mathrm{d}T. \tag{1.5}$$

A third, equivalent property is the symmetry of the mixed second derivatives:

$$\frac{\partial^2 U}{\partial V\, \partial T} = \frac{\partial^2 U}{\partial T\, \partial V}. \tag{1.6}$$

Under the usual smoothness and domain assumptions, these three statements—path independence, vanishing integrals around closed paths, and symmetry of the mixed derivatives—are mathematically equivalent. Their equivalence follows from Green's theorem, the two-dimensional form of Stokes' theorem. We will not explore the mathematical details here.

But Green's theorem for a closed contour C and domain enclosed by the contour D is given by,

$$\oint_C (L\,\mathrm{d}x + M\,\mathrm{d}y) = \iint_D \left(\frac{\partial M}{\partial x} - \frac{\partial L}{\partial y}\right) \mathrm{d}A \tag{1.7}$$

for our case,

$$\oint_C \mathrm{d}U = \oint_C \left[\left(\frac{\partial U}{\partial V} \right)_T \mathrm{d}V + \left(\frac{\partial U}{\partial T} \right)_V \mathrm{d}T \right] = \iint_D \left(\frac{\partial^2 U}{\partial V \partial T} - \frac{\partial^2 U}{\partial T \partial V} \right) \mathrm{d}A = 0 \tag{1.8}$$

where the first equality requires U to have exact differential and second equality is Green's theorem and third equality require the symmetry in the mixed second derivative.

Heat and work are not state functions: their values depend on the path taken between two states. They are therefore called **path functions**, and their inexact differentials are denoted by $đq$ and $đw$. The differential form of the first law is

$$\boxed{\mathrm{d}U = đq + đw}. \tag{1.9}$$

This is the most useful form of the first law. Since $\mathrm{d}U$ is exact,

$$\oint (đq + đw) = 0. \tag{1.10}$$

Thus, although heat and work are individually path dependent, their sum gives the change in the state function U.

We now consider expansion work. Suppose a gas is confined in a cylinder by a movable piston of area A. If the gas expands and moves the piston by a small distance $\mathrm{d}z$ against an external pressure p_{ex}, the work done on the gas is

$$đw = -F\,\mathrm{d}z = -p_{\mathrm{ex}} A\,\mathrm{d}z = -p_{\mathrm{ex}}\,\mathrm{d}V. \tag{1.11}$$

The negative sign reflects our convention that work done *on* the system is positive: during expansion, the gas does work on the surroundings, so $w < 0$.

If the external pressure is constant,

$$w = -\int_{V_1}^{V_2} p_{\mathrm{ex}}\,\mathrm{d}V = -p_{\mathrm{ex}}(V_2 - V_1) = -p_{\mathrm{ex}}\Delta V. \tag{1.12}$$

For free expansion into a vacuum, $p_{\mathrm{ex}} = 0$, and hence

$$w = 0. \tag{1.13}$$

In a reversible expansion, the gas pressure exceeds the external pressure by only an infinitesimal amount. In the reversible limit, $p_{\mathrm{ex}} \to p$, so

$$w_{\mathrm{rev}} = -\int_{V_1}^{V_2} p\,\mathrm{d}V. \tag{1.14}$$

The condition $p = p_{\mathrm{ex}}$ is called **mechanical equilibrium**; a reversible expansion proceeds through states infinitesimally close to this condition.

For an ideal gas undergoing a reversible isothermal expansion, $p = nRT/V$ and T is constant. Therefore,

$$\begin{aligned} w_{\mathrm{rev}} &= -\int_{V_1}^{V_2} p\,\mathrm{d}V \\ &= -\int_{V_1}^{V_2} \frac{nRT}{V}\,\mathrm{d}V \\ &= -nRT \ln\left(\frac{V_2}{V_1}\right). \end{aligned} \tag{1.15}$$

An ideal gas has an internal energy that depends only on temperature. Thus, for any isothermal process,

$$\Delta U = 0, \tag{1.16}$$

and the first law gives $q = -w$. For an irreversible expansion against a constant external pressure and a reversible expansion between the same states, respectively,

$$q_{\text{irr}} = p_{\text{ex}} \Delta V \tag{1.17}$$

and

$$q_{\text{rev}} = nRT \ln \left(\frac{V_2}{V_1} \right). \tag{1.18}$$

The two paths have the same initial and final states, but different values of q and w, confirming that heat and work are path dependent.

Among expansion paths connecting the same equilibrium states, the reversible path produces the maximum work: with our sign convention, w_{rev} is the most negative. For an isothermal ideal-gas expansion, the heat absorbed by the gas equals the work it does on the surroundings, while its internal energy remains unchanged. Clearly, at constant volume, no work is done, and we have,

$$\mathrm{d}U = đq_V \tag{1.19}$$

and,

$$0 = \oint đq_V \tag{1.20}$$

where subscript V denotes constant volume (isochoric) process. Or equivalently, for non-infinitesimal change,

$$\Delta U = q_V \tag{1.21}$$

a natural question this raises, is that whether there is a state function, that is the same as heat added at constant pressure? To answer this, consider enthalpy, which is defined as,

$$H \equiv U + pV \tag{1.22}$$

then take differential on both sides, we have,

$$\begin{aligned}
\mathrm{d}H &= \mathrm{d}U + \mathrm{d}(pV) \\
&= \mathrm{d}U + V\mathrm{d}p + p\mathrm{d}V \\
&= đq + đw + V\mathrm{d}p + p\mathrm{d}V \\
&= đq - p_{\text{ex}}\mathrm{d}V + V\mathrm{d}p + p\mathrm{d}V \\
&= đq - p\mathrm{d}V + V\mathrm{d}p + p\mathrm{d}V \\
&= đq + V\mathrm{d}p \\
&= đq_p
\end{aligned} \tag{1.23}$$

where we have assumed system is in mechanical equilibrium with surrounding from line 4 to 5. Thus, enthalpy is heat added to the system at constant pressure.

$$\mathrm{d}H = đq_p \tag{1.24}$$

enthalpy is a state function, since internal energy, pressure and volume are state functions. So we have,

$$0 = \oint \mathrm{d}H = \oint đq_p \tag{1.25}$$

We now turn to the experimental measurement of internal energy and enthalpy. In quantum mechanics, the solutions of the time-independent Schrödinger equations are energy levels (and stationary states), which are tested using **spectroscopy**: photons are absorbed or emitted when their frequencies match the energy gaps between quantum states. The corresponding experimental technique in thermodynamics is **calorimetry**, which determines heat transfer from the resulting change in temperature.

The central experimental quantity is the **heat capacity**, which relates the heat transferred to the temperature change. There are two types of heat capacities, constant volume heat capacity C_V and constant pressure heat capacity C_p, defined as follows

$$C_V \equiv \left(\frac{\partial U}{\partial T}\right)_V = \frac{đq_V}{\mathrm{d}T} \tag{1.26}$$

and

$$C_p \equiv \left(\frac{\partial H}{\partial T}\right)_p = \frac{đq_p}{\mathrm{d}T}, \tag{1.27}$$

For a small temperature interval over which the heat capacity is approximately constant,

$$\delta T \simeq \frac{q_V}{C_V} \tag{1.28}$$

at constant volume, and

$$\delta T \simeq \frac{q_p}{C_p} \tag{1.29}$$

at constant pressure. Thus, a substance with a large heat capacity requires a large amount of heat to produce a small temperature increase, whereas a substance with a small heat capacity changes temperature more readily.

In general, heat capacity depends on temperature. At constant volume,

$$U(T_2, V) - U(T_1, V) = \int_{T_1}^{T_2} C_V(T)\,\mathrm{d}T, \tag{1.30}$$

or equivalently,

$$U(T_2, V) = U(T_1, V) + \int_{T_1}^{T_2} C_V(T)\,\mathrm{d}T. \tag{1.31}$$

Similarly, at constant pressure,

$$H(T_2, p) = H(T_1, p) + \int_{T_1}^{T_2} C_p(T)\,\mathrm{d}T. \tag{1.32}$$

Therefore, if the internal energy or enthalpy is known at a reference temperature, calorimetric measurements of $C_V(T)$ or $C_p(T)$ allow us to determine its value at any other temperature.

We have already considered an isothermal process, which occurs at constant temperature. We now turn to another important type of process: an **adiabatic process**, in which no heat is exchanged between the system and its surroundings.

Consider a reversible adiabatic process involving one mole of a monatomic ideal gas, taken from an initial state (V_1, T_1) to a final state (V_2, T_2). Its constant-volume heat capacity is

$$C_V = \frac{3}{2}R. \tag{1.33}$$

Because the process is adiabatic, $đq = 0$. For a reversible process, $p_{\text{ex}} \to p$, so the first law gives

$$\mathrm{d}U = đw = -p\,\mathrm{d}V = -\frac{RT}{V}\,\mathrm{d}V, \tag{1.34}$$

where we have used the ideal-gas equation for one mole.

The internal energy of an ideal gas depends only on temperature, so

$$\mathrm{d}U = C_V\,\mathrm{d}T. \tag{1.35}$$

Equating the two expressions for $\mathrm{d}U$ gives

$$C_V\,\mathrm{d}T = -\frac{RT}{V}\,\mathrm{d}V. \tag{1.36}$$

Using $C_V = 3R/2$ and rearranging,

$$\frac{3}{2}\frac{\mathrm{d}T}{T} = -\frac{\mathrm{d}V}{V}. \tag{1.37}$$

Integrating between the initial and final states,

$$\frac{3}{2}\int_{T_1}^{T_2}\frac{\mathrm{d}T}{T} = -\int_{V_1}^{V_2}\frac{\mathrm{d}V}{V}, \tag{1.38}$$

which gives

$$\frac{3}{2}\ln\left(\frac{T_2}{T_1}\right) = -\ln\left(\frac{V_2}{V_1}\right). \tag{1.39}$$

Therefore,

$$\left(\frac{T_2}{T_1}\right)^{3/2} = \frac{V_1}{V_2}, \tag{1.40}$$

This relation allows any one of V_1, T_1, V_2, and T_2 to be determined from the other three.

During an isothermal expansion, the gas absorbs heat and converts it into work while its internal energy remains unchanged. During an adiabatic expansion, no heat enters the gas. It must therefore do work at the expense of its own internal energy, causing its temperature to decrease. Thus if $T_2 > T_1$ then we must have $V_2 < V_1$ and vice versa.

In this chapter, we introduced the first law of thermodynamics, together with several basic thermodynamic concepts and processes. The first law imposes a fundamental constraint: any physically possible process must conserve energy. However, energy conservation alone does not tell us whether a process will actually occur or in which direction it will proceed.

In the next chapter, we will introduce the **second law of thermodynamics**. The second law determines the direction of spontaneous change and places fundamental limits on the efficiency of heat engines—questions that played a central role in the development of thermodynamics during the Industrial Revolution.

2 The Second Law

The first law tells us which processes are *permissible*: those that conserve energy. It does not tell us which of them actually happen. A cup of tea left on a desk goes cold, but a cold cup never draws heat back out of the room and reheats itself — and yet energy would be conserved either way. To run the process backwards you need a kettle, and the kettle has to be plugged into the socket: the reverse direction must be driven by external work.

The second law supplies what is missing. A **spontaneous** process is one that proceeds without being driven by external work, and the second law tells us which processes these are. In this chapter we state the second law, define entropy, and verify that $\mathrm{d}S$ is an exact differential. (or equivalently, entropy is a state function)

We then connect entropy to experiment through heat capacities, fix its absolute scale using the third law, and finally apply it to the Carnot cycle, where it places a ceiling on the efficiency of every heat engine — the question that drove the subject during the Industrial Revolution.

The first law introduces a state function internal energy U, by combining two path functions q and w. The second law, introduces another state function S - Entropy. The second law can be stated as: *A spontaneous process in an isolated system is accompanied by an increase in entropy.* Note this does not mean entropy of the system must increase, because it could decrease, provided that entropy of surrounding is a even bigger increase.

We stated the second law in words, but we haven't defined what entropy is, and haven't written down a maths expression for it, lets do that now. The most precise statement of the second law is the Clausius inequality,

$$\mathrm{d}S \geq \frac{đq}{T} \tag{2.1}$$

where $\mathrm{d}S$ means entropy increase of the system, and the equality holds for reversible process, and it serves as a definition of entropy,

$$\mathrm{d}S \equiv \frac{đq_{\mathrm{rev}}}{T} \tag{2.2}$$

which is heat added reversibly at temperature T. Analogous to work done reversibly at mechanical equilibrium, resersible heat transfer occurs at thermal equilibrium. The Clausius inequality clearly implies for an isolated system, such as the whole universe, we have,

$$\mathrm{d}S \geq 0 \tag{2.3}$$

Entropy is a state function so we have,

$$\oint \mathrm{d}S = \oint \frac{đq_{\mathrm{rev}}}{T} = 0 \tag{2.4}$$

and

$$\oint \frac{đq}{T} \leq 0 \tag{2.5}$$

thus the second law also relate a path function to a state function. However, it is not obviously at all why dividing reversible incremental heat transfer by temperature converts it into an exact differential, lets look at this more closely. We now show entropy is a state function in the case of an ideal gas. Turns out even through we wrote $U \equiv U(V,T)$ for

an ideal gas, the internal energy is actually just a function of temperature, $U \equiv U(T)$, we have,

$$\mathrm{d}U = C_V(T)\mathrm{d}T \tag{2.6}$$

now consider first law for reversible process,

$$\mathrm{d}U = đq_{\text{rev}} + đw_{\text{rev}} = đq_{\text{rev}} - p\mathrm{d}V \tag{2.7}$$

we used the reversible work done expression here. The combine the two equations and rearrange,

$$đq_{\text{rev}} = C_V(T)\mathrm{d}T + p\mathrm{d}V \tag{2.8}$$

divide by T on both sides,

$$\frac{đq_{\text{rev}}}{T} = \frac{C_V(T)}{T}\mathrm{d}T + \frac{p}{T}\mathrm{d}V \tag{2.9}$$

use ideal gas equation,

$$\mathrm{d}S = \frac{đq_{\text{rev}}}{T} = \frac{C_V(T)}{T}\mathrm{d}T + \frac{nR}{V}\mathrm{d}V \tag{2.10}$$

clearly we also have,

$$\mathrm{d}S = \left(\frac{\partial S}{\partial T}\right)_V \mathrm{d}T + \left(\frac{\partial S}{\partial V}\right)_T \mathrm{d}V \tag{2.11}$$

equating coefficients yields,

$$\left(\frac{\partial S}{\partial T}\right)_V = \frac{C_V(T)}{T}, \qquad \left(\frac{\partial S}{\partial V}\right)_T = \frac{nR}{V} \tag{2.12}$$

then clearly,

$$\frac{\partial^2 S}{\partial V \partial T} = 0 = \frac{\partial^2 S}{\partial T \partial V} \tag{2.13}$$

thus $\mathrm{d}S$ is indeed an exact differential in the case for a mole of monotonic ideal gas, and entropy is a state function for ideal gas. You could show $đq_{\text{rev}}$ is not an exact differential using similar method. Now we show entropy is a state function for any arbitrary substance A, lets denote entropy of ideal gas (ig), then consider an isolated system only two parts: ideal gas, and material A, in thermal equilibrium with eachother at common temperature T, so we have,

$$\mathrm{d}S_{\text{A}} \equiv \frac{đq_{\text{A, rev}}}{T} = \frac{đq_{\text{ig, rev}}}{T} \equiv \mathrm{d}S_{\text{ig}} \tag{2.14}$$

clearly, since $\mathrm{d}S_{\text{ig}}$ is an exact differential, $\mathrm{d}S_{\text{A}}$ is also an exact differential, thus entropy for any arbitrary material is a state function. We haven't mentioned this, but the mathematical reason behind the equivilence between integral on a closed contour vanishing. We now make a connection to experiment, we talked about how to predict internal energy and enthalpy change for a given temperature range using heat capacities, now we talk about how to use heat capacities to calculate entropy change for a given temperature

range. Let's consider constant volume first,

$$\begin{aligned}\int_{V,T_1}^{V,T_2} \mathrm{d}S &= \int_{V,T_1}^{V,T_2} \frac{đq_{\text{rev, v}}}{T} \\ &= \int_{V,T_1}^{V,T_2} \frac{\mathrm{d}U}{T} \\ &= \int_{V,T_1}^{V,T_2} \left(\frac{\partial U}{\partial T}\right)_V \frac{\mathrm{d}T}{T} \\ &= \int_{V,T_1}^{V,T_2} \frac{C_V(T)\mathrm{d}T}{T}\end{aligned} \tag{2.15}$$

Thus,

$$S(V,T_2) = S(V,T_1) + \int_{T_1}^{T_2} \frac{C_V(T)\mathrm{d}T}{T} \tag{2.16}$$

and similarly for constant pressure,

$$S(p,T_2) = S(p,T_1) + \int_{T_1}^{T_2} \frac{C_p(T)\mathrm{d}T}{T} \tag{2.17}$$

these results even holds for non-reversible process and non-ideal gas. Energy and enthalpy do not have absolute scale, as the first law never actually defined U but it is written in terms of difference ΔU or $\mathrm{d}U$. However, entropy do have an absolute scale, and this scale is provided by the third law of thermodynamics, which states: *entropy for all crystalline substance goes to zero as temperature goes to zero*,

$$\lim_{T\to 0} S = 0 \tag{2.18}$$

And since we have equations gives us temperature dependence of entropy, provided that we have the heat capacities at the temperature range, we can calculate the absolute entropy of any crystal. Actually, there is also the zeroth law of thermodynamics, which defines thermodynamic temperature. At the end lets show the entropy change for any closed cycle is zero, and make a connection to applications to heat engines.

Consider a Carnot cycle, which is made of four individual processes, a quantity with bigger number in the subscript means the larger quantity i.e. $(T_2 > T_1)$ and $V_4 > V_3 > V_2 > V_1$

1. Isothermal, reversible expansion $(T_2, V_1) \to (T_2, V_3)$
2. Adiabatic, reversible expansion $(T_2, V_3) \to (T_1, V_4)$
3. Isothermal, reversible compression $(T_1, V_4) \to (T_1, V_2)$
4. Adiabatic, reversible compression $(T_1, V_2) \to (T_2, V_1)$ (2.19)

Which ends up where we started at, i.e. a closed cycle. We can first check whether internal energy change is zero for this cycle. Clearly, isothermal process do not change energy, and the two adiabatic processes, give internal energy change of

$$\Delta U_2 = C_V(T_1 - T_2) = -\Delta U_4 \tag{2.20}$$

thus internal energy change does indeed vanish for this cycle. Now, entropy change, clearly, reversible adiabatic process does not change entropy, so we only need to look at

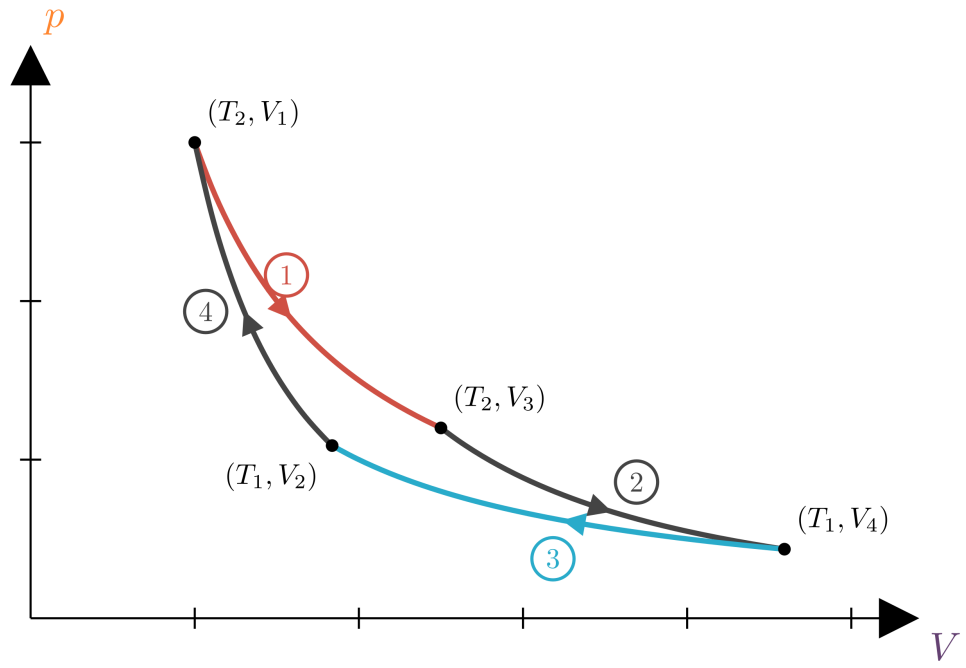


Figure 1: The Carnot cycle in the p–V plane. Processes 1 and 3 are isothermal, at T_2 and T_1 respectively; processes 2 and 4 are adiabatic. Arrows show the direction of traversal.

entropy change for isothermal reversible process, which is in general

$$\Delta S = \int_{T,V_1}^{T,V_2} \frac{đq}{T} = \frac{1}{T}\int_{T,V_1}^{T,V_2} đq_{\text{rev}} = \frac{q_{\text{rev}}}{T} \tag{2.21}$$

But we know heat transferred for a isothermal, reversible expansion from Eq. (1.13), thus, the entropy change for an isothermal, reversible expansion is

$$\Delta S = nR\ln\frac{V_2}{V_1} \tag{2.22}$$

Subtlety, although we derived this by considering a reversible expansion, this expression holds more generally for non-reversible isothermal expansion, because entropy is a state function. The equation says, entropy gets bigger as volume gets bigger. Thus for process 1 and 3 in Carnot cycle, we have,

$$\Delta S_1 = R\ln\frac{V_3}{V_1}, \qquad \Delta S_3 = R\ln\frac{V_2}{V_4}, \tag{2.23}$$

so the entropy change for the whole cycle is,

$$\Delta S_1 + \Delta S_3 = R\ln\frac{V_3}{V_1} + R\ln\frac{V_2}{V_4} = R\ln\frac{V_3V_2}{V_1V_4} \tag{2.24}$$

but we know know for an adiabatic reversible process for a monotomic ideal gas Eq. (1.35), in our case, this becomes,

$$\left(\frac{T_1}{T_2}\right)^{3/2} = \frac{V_3}{V_4}, \qquad \left(\frac{T_2}{T_1}\right)^{3/2} = \frac{V_2}{V_1}, \tag{2.25}$$

substituting these results back yields,

$$\Delta S_1 + \Delta S_3 = R \ln \left(\frac{T_2}{T_1}\right)^{3/2} \left(\frac{T_1}{T_2}\right)^{3/2} = R \ln 1 = 0 \tag{2.26}$$

Thus entropy change is indeed zero for a mole of monoatomic gas, in a Carnot cycle. For non Carnot cycles, we dissect it into many many smaller Carnot cycles, since, the internal paths cancel, entropy vanishes for an arbitrary cycle. This is again reminiscent of Green's theorem, which related many little circulation to a big overall circulation.

In fact, the cycle above is can be regarded as a heat engine, which absorbs heat q_1 from hot bath at temperature T_2 in process 1, dumps this heat into cold sink with temperature T_1 in process 3, outputs work $-w$ to the surrounding in a full cycle. We can define the efficiency of this ideal gas reversible heat engine as a ratio of work done on surrounding and heat absorbed from surrounding in a cycle,

$$\eta \equiv \frac{\text{work done on surroundings}}{\text{heat absorbed from hot bath}} \tag{2.27}$$

then,

$$\eta_{\text{rev}} = \frac{-w}{q_1} = \frac{q}{q_1} = \frac{q_1 + q_3}{q_1} = 1 + \frac{q_3}{q_1} \tag{2.28}$$

then from the cycle we have,

$$0 = \oint \mathrm{d}S = \frac{q_1}{T_2} + \frac{q_3}{T_1} \tag{2.29}$$

a slight rearangement gives,

$$\frac{q_3}{q_1} = -\frac{T_1}{T_2} \tag{2.30}$$

thus the efficiency this Carnot heat engines is,

$$\eta_{\text{rev}} = 1 - \frac{T_1}{T_2} \tag{2.31}$$

So the engine has an efficiency of 100 % if the cold sink is a absolute zero, or the hot bath at infinite temperature, both are not practical. The heat engine has an efficient of 0, if the cold sink and hot bath is at the same temperature. The Carnot engine is the most efficient heat engine which is theoretically possible.

This gives us a criterion for spontaneous change, but not a convenient one. The Clausius inequality contains $đq$, and heat is a path function: it depends on how a process is carried out, not merely on where the system begins and ends. And the clean statement $\mathrm{d}S \geq 0$ holds for an isolated system, whereas nothing in a laboratory is isolated.

What we actually have is a flask held at a fixed temperature, at either fixed volume or fixed pressure. In the next chapter we rearrange the Clausius inequality to suit those conditions, and obtain the free energies: state functions of the system alone whose decrease marks a spontaneous change. They contain no new physics — they are the second law written in a form we can take into the lab.

3 Free Energy

The second law tells us which processes are spontaneous, but not in a form that is easy to use. The Clausius inequality contains the heat, and heat depends on the path taken; and its tidiest statement, $\mathrm{d}S \geq 0$, applies to an isolated system, which is not what sits on a laboratory bench.

This chapter turns it into something practical. Rearranging the Clausius inequality for the conditions we can actually impose — a fixed temperature, together with either a fixed volume or a fixed pressure — produces two new state functions: the Helmholtz free energy A and the Gibbs free energy G. Both say exactly what the second law said; the difference is that they say it in the variables we control. From them follow the fundamental equations, the Maxwell relations, which trade quantities that are hard to measure for quantities that are easy, and in the end a quantitative handle on the yield of a chemical reaction.

Rearrange the Clausius inequality gives,

$$đq \leq T\mathrm{d}S \tag{3.1}$$

Thus,

$$\mathrm{d}U = đq + đw \leq T\mathrm{d}S + đw = T\mathrm{d}S - p_{\mathrm{ex}}\mathrm{d}V \tag{3.2}$$

thus at constant entropy and volume, a spontaneous process decreases internal energy of the system,

$$\mathrm{d}U_{S,V} \leq 0 \tag{3.3}$$

the equality holds at equilibrium. This is neat, but not very useful, because it is quite difficult to keep entropy of the system constant in the lab. We now define the Helmholtz free energy as,

$$A \equiv U - TS \tag{3.4}$$

then,

$$\mathrm{d}A = \mathrm{d}U - T\mathrm{d}S - S\mathrm{d}T \leq T\mathrm{d}S - p_{\mathrm{ex}}\mathrm{d}V - T\mathrm{d}S - S\mathrm{d}T = -p_{\mathrm{ex}}\mathrm{d}V - S\mathrm{d}T \tag{3.5}$$

thus at constant volume and temperature,

$$\mathrm{d}A_{V,T} \leq 0 \tag{3.6}$$

where equality holds for equilibrium. So if the system is at constant temperature (much easier than keeping the entropy constant) and volume, then a spontaneous process lowers the Helmholtz free energy. This is in fact a masked statement of second law, since we used the Clausius inequality to derive these relations. It also quite common to keep the pressure of the system constant. We now define the Gibbs free energy,

$$G \equiv U + pV - TS = A + pV = H - TS \tag{3.7}$$

then following similar argument we have,

$$\mathrm{d}G = \mathrm{d}A + p\mathrm{d}V + V\mathrm{d}p \leq -p_{\mathrm{ex}}\mathrm{d}V - S\mathrm{d}T + p\mathrm{d}V + V\mathrm{d}p = V\mathrm{d}p - S\mathrm{d}T \tag{3.8}$$

thus at constant pressure at temperature, we have,

$$\mathrm{d}G_{p,T} \leq 0 \tag{3.9}$$

where equality holds for equilibrium. This is a very useful condition indeed. Now consider for a reversible process (i.e. the system is in both thermal equilibrium and mechanical equilibrium with surrounding) we have:

$$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V \tag{3.10}$$

this known as a the fundamental equation. Since $\mathrm{d}U$ is an exact differential, we may also have,

$$\mathrm{d}U = \left(\frac{\partial U}{\partial S}\right)_V \mathrm{d}S + \left(\frac{\partial U}{\partial V}\right)_S \mathrm{d}V \tag{3.11}$$

by comparing coefficients of $\mathrm{d}V$, we have,

$$p = -\left(\frac{\partial U}{\partial V}\right)_S \tag{3.12}$$

and,

$$T = \left(\frac{\partial U}{\partial S}\right)_V \tag{3.13}$$

the last relation is a compact expression relating temperature (zeroth law concept), internal energy (first law concept), and entropy (second law concept). Similarly,

$$\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p \tag{3.14}$$

and,

$$\mathrm{d}A = -p\mathrm{d}V - S\mathrm{d}T \tag{3.15}$$

and,

$$\mathrm{d}G = V\mathrm{d}p - S\mathrm{d}T \tag{3.16}$$

Subtlety, fundamental equations holds for non-reversible processes, since internal energy, enthalpy and the free energies are state functions, but earlier, we had for a non-reversible spontaneous process, it must be inequalities. This is because, the earlier expression hols for non-reversible process between two non-equilibrium states. Whereas, the fundamental equations holds for non-reversible process between two equilibrium state points. For example, earlier equations applies to water below zero Celsius, it spontaneously freezes, lowering its Gibbs free energy at constant pressure and temperature. Fundamental equation does not apply in this case, since water is not in its equilibrium state below zero Celsius, the equilibrium state is ice.

Now we focus on Gibbs energy. Clearly, from coefficients, we have,

$$V = \left(\frac{\partial G}{\partial p}\right)_T, \qquad S = -\left(\frac{\partial G}{\partial T}\right)_p, \tag{3.17}$$

but G is a state function, thus, we have,

$$\oint \mathrm{d}G = 0, \tag{3.18}$$

and,

$$\frac{\partial^2 G}{\partial T \partial p} = \frac{\partial^2 G}{\partial p \partial T} \tag{3.19}$$

then,

$$\left(\frac{\partial V}{\partial T}\right)_p = \left(\frac{\partial S}{\partial p}\right)_T \tag{3.20}$$

this is known as a maxwell relation. So if we can measure how volume changes if we change temperature at constant temperature, which is easy to measure, we also directly know how entropy changes as we vary pressure at constant temperature, which would be hard to measure directly. Then similarly, from the fundamental equation of Helmholtz free energy we have,

$$\left(\frac{\partial p}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T \tag{3.21}$$

and from enthalpy,

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p \tag{3.22}$$

and from internal energy,

$$\left(\frac{\partial T}{\partial V}\right)_S = \left(\frac{\partial p}{\partial S}\right)_V \tag{3.23}$$

We now turn to pressure dependence of G, for an ideal gas,

$$\begin{aligned}
\int_{p_1,T}^{p_2,T} \mathrm{d}G &= \int_{p_1,T}^{p_2,T} \left(\frac{\partial G}{\partial p}\right)_T \mathrm{d}p \\
&= \int_{p_1,T}^{p_2,T} V \mathrm{d}p \\
&= \int_{p_1,T}^{p_2,T} \frac{nRT}{p} \mathrm{d}p \\
&= nRT \int_{p_1,T}^{p_2,T} \frac{\mathrm{d}p}{p} \\
&= nRT \ln \frac{p_2}{p_1}
\end{aligned} \tag{3.24}$$

thus,

$$G(p_2, T) = G(p_1, T) + nRT \ln \frac{p_2}{p_1} \tag{3.25}$$

Now we look at the temperature dependence of Gibbs free energy, which is negative entropy, however, there is a more convenient form. Consider,

$$\left(\frac{\partial (G/T)}{\partial T}\right)_p = \frac{1}{T}\left(\frac{\partial G}{\partial T}\right)_p - \frac{G}{T^2} = -\frac{S}{T} - \frac{G}{T^2} = -\frac{TS + G}{T^2} = -\frac{H}{T^2} \tag{3.26}$$

which is the Gibbs-Helmholtz relation. To illustrate why it is useful, we will state an important result, that we will show much later.

At the end, we consider a simple but important chemical reaction known as the Harber process, which converts hydrogen and nitrogen into ammonia,

$$\mathrm{N_2} + 3\mathrm{H_2} \rightleftharpoons 2\mathrm{NH_3} \qquad \Delta H^{\ominus}_{298\,\mathrm{K}} = -92.28\,\mathrm{kJ\,mol^{-1}_{N_2}}. \tag{3.27}$$

this process uses 2% of the total energy of the world, and feeds 4 Billion people, which is about half the human population, so it is important that we do this reaction cheaply and

efficiently. The $\ominus$ symbol denotes standard state, which mean two things: 1. The substance is pure 2. pressure of 1 bar, which is very very close to the atmospherical pressure. $1\,\mathrm{atm} = 1.01325\,\mathrm{bar}$. The Δ_r denotes the total amount of this quantity on the right-hand side minus the total amount on the left-hand side, weighed by the stoichiometric coefficients. Then,

$$\Delta_\mathrm{r}H^\ominus(T) = 2H_m^\ominus(\mathrm{NH_3};T) - 3H_m^\ominus(\mathrm{H_2};T) - H_m^\ominus(\mathrm{N_2};T) \tag{3.28}$$

where the subscript m indicates a mole of this substance. We now introduce the stoichiometric coefficients, so that,

$$\Delta_\mathrm{r}H^\ominus(T) = \sum_J \nu_J H_m^\ominus(J;T) \tag{3.29}$$

and similarly for Gibbs energy,

$$\Delta_\mathrm{r}G^\ominus(T) = \sum_J \nu_J G_m^\ominus(J;T) \tag{3.30}$$

where J denotes the chemical species. We define the equilibrium constant K_p as,

$$K_p \equiv \left[\prod_J \left(\frac{p_J}{p^\ominus}\right)^{\nu_J}\right]_\mathrm{eqm} \tag{3.31}$$

For a gas phase chemical reaction, there is a well-known relation between standard Gibbs energy of reaction and equilibrium constant,

$$\Delta_\mathrm{r}G^\ominus = -RT\ln K_p \tag{3.32}$$

substituting the equilibrium constant relation into Gibbs-Helmholtz relation yields,

$$\frac{\mathrm{d}\ln \mathrm{K_p}}{\mathrm{d}T} = \frac{\Delta_\mathrm{r}H^\ominus}{RT^2} \tag{3.33}$$

which is known as **Van't Hoff equation** and gives us the equilibrium constant's temperature dependence. For example, if the reaction is exothermic as $\Delta_\mathrm{r}H^\ominus < 0$, K_p decreases if you increase temperature, i.e. it moves to the endothermic side. This is in fact Le Chatelier's principle, (i.e. the system moves in an opposite direction in order to counteract the change) Thermodynamics puts this into a quantitative basis, and allows us to calculate the exact equilibrium constant for a given temperature, thus predicting the yield.

We started with a system described by only a few macroscopic variables, and with three statements about it. The first law tells us which processes are allowed: those that conserve energy. The second law tells us which of the allowed processes actually happen, and in which direction. The third law sets the zero point of the entropy scale, so that we can tabulate absolute entropies and not only differences. Every other result in these notes follows from these three.

The work of deriving those results is worth doing. It gave us an upper limit on the efficiency of any heat engine, even though we know nothing about how a particular engine is built. It also gave us the temperature dependence of an equilibrium constant, and therefore the yield of the Haber process, even though we know nothing about how nitrogen

and hydrogen actually react. In both cases the question seems to require a detailed mechanism. In fact it requires only a state function and a constraint.

This is the strength of thermodynamics, and also its limit. At no point did we need a model of what matter is made of. This is why thermodynamics was already useful to engineers decades before scientists agreed that atoms exist. But there is a cost: thermodynamics tells us what must happen, not why it happens. It does not explain why entropy increases, or why a heat capacity has the value it has. To answer such questions we must count microscopic states, and that is **statistical mechanics**.